



# INTRODUCTION TO SQL

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# What is SQL?

SQL stands for **Structured Query Language**. It's a programming language designed specifically for managing and retrieving data from databases.

Think of SQL as a way to have a conversation with a database you ask questions, and it gives you answers



# Core Concepts: The Building Blocks

**01.**

## Database

A database is like a digital filing cabinet. It's an organized collection of data stored on a computer. Instead of physical files in folders, data in a database is organized into tables

**02.**

## Table

A table is like a spreadsheet. It's the actual structure where data lives. Each database can contain many tables, and each table stores a specific type of information

**03.**

## Rows and Columns

Tables are made up of rows and columns, just like a spreadsheet;

- \* Columns ("fields"): The categories of information. Each has a name and represents one attribute.

- \* Rows ("records"): Individual entries of data. Each represents one complete record.

# Types of Databases

## Relational Databases

**Structure:** Tables with rows and columns

**Examples:** MySQL, PostgreSQL, SQLite, Oracle

## NoSQL Databases

**Structure:** Flexible key-value, document, column, or graph

**Examples:** MongoDB (document), Redis (key-value), Cassandra (column)

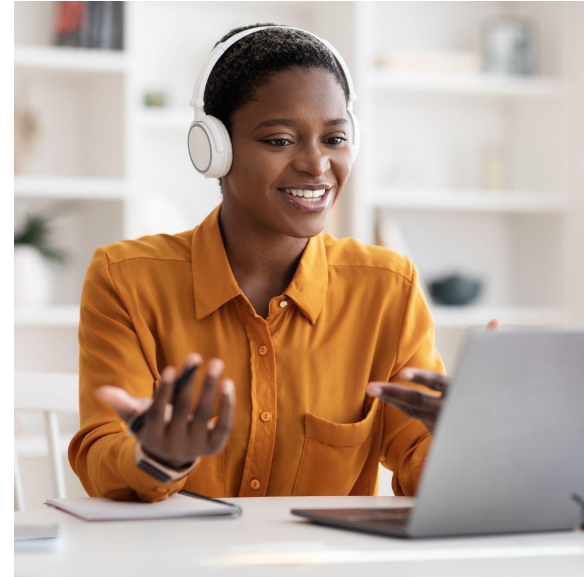
## Cloud Databases

**Structure:** Can be relational or NoSQL, hosted in the cloud

**Examples:** Amazon RDS, Google BigQuery, Azure SQL

# Why SQL Matters

1. Universal skill: Nearly every company that works with data uses SQL. Learning it makes you valuable across industries.
2. Efficient data retrieval: Instead of manually looking through thousands of records, SQL lets you extract exactly what you need in seconds.
3. Data analysis: As a data scientist, you'll use SQL constantly to explore data before building models.



# WHAT CAN SQL DO?

SQL lets you manipulate databases

**01.**

SQL retrieve data from a database

**02.**

SQL insert records in a database

**03.**

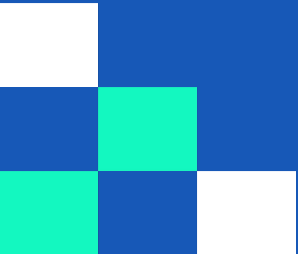
SQL can delete records from a database

**04.**

SQL can update records in a database

**05.**

SQL can create new databases



# Basic SQL Commands

**SELECT** – chooses the columns you want to see in the result.

**FROM** – specifies the table where the data is stored.

**WHERE** – filters rows based on conditions you define.

**LIMIT** – restricts the number of rows returned.

**ORDER BY** – sorts the result set by one or more columns.

**DISTINCT** – removes duplicate values from the result.

**COUNT()** – returns the number of rows that match a query.

**SUM()** – adds up values in a numeric column.

**AVG()** – computes the average of a numeric column.

**MIN()** – returns the smallest value in a column.

**MAX()** – returns the largest value in a column.

**GROUP BY** – groups rows so aggregate functions can be applied per category.



**THANK YOU**